

Existence of Invariants in Periods of Generalized Collatz Sequences

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Abstract—A generalization of Collatz sequence and discuss patterns in the terms of periodic Collatz sequences is considered. These patterns can also be interpreted as some regularities in the digits of p-adic representations of these terms. These patterns were explored using the algorithm of division by two in any odd based numeral system. In the paper an implementation of this algorithm as a Pascal program for ternary numeral system was given.

Keywords—Collatz conjecture, p -adic numbers, cycles, rational numbers

I. INTRODUCTION

There are many generalizations of Collatz conjecture [7]. In the classical case there are only 2 functions: $3x + 1$ and $x/2$. By applying these functions recursively to any initial number if the number is odd and even, respectively, one obtains a sequence called as Collatz sequence. Collatz conjecture claims that irrespective of what the initial number is, the Collatz sequence always contains 1. Another interesting fact is that 1 is a term of periodic Collatz sequence consisted of only numbers 1, 4, 2. It is also not proven if this is the only possible period of the Collatz sequence. In the current paper we generalize the Collatz sequence by using functions of type $px + k$ and x/q , where $k \in \mathbb{Z}$ and $(p, q) = 1$. We show that as in the classical case there are some fixed numbers a, b , which we call invariants, that are not dependent on the chosen permutation of the functions determining cycle.

II. FUNCTIONS, CYCLES AND SPECIAL NUMBERS

Let $B_i \in \left\{ T(x) = \frac{x}{q} \right\} \cup \left\{ S_k(x) = \frac{px+k}{q}, k \in \mathbb{Z} \right\}, i = \overline{0, n+m-1}$, where $p, q > 1$ are relatively prime number. Suppose that $P = B_0 \circ B_1 \circ \dots \circ B_{n+m-1}$. Assume that T is repeated n times and S_k is repeated m times in P . Let $m > 0$ and $n \geq 0$. Let B_i and B_j be the same if i and j are equal mod $n + m$:

$$P = \dots B_0 \circ B_1 \circ \dots \circ B_{n+m-1} \circ B_0 \circ B_1 \circ \dots \circ B_{n+m-1} \circ B_0 \circ B_1 \circ \dots \circ B_{n+m-1} \circ \dots$$

Consider the solution x_0 of the equation $B_0 \circ B_1 \circ \dots \circ B_{n+m-2} \circ B_{n+m-1}(x) = x$. Similarly, x_1 is the solution of $B_1 \circ \dots \circ B_{n+m-2} \circ B_{n+m-1} \circ B_0(x) = x$ ($x_0 = B_0(x_1)$). Analogously for x_i . Finally, the solution of $B_{n+m-1} \circ B_0 \circ B_1 \circ \dots \circ B_{n+m-2}(x) = x$ ($x_{n+m-1} = B_{n+m-1}(x_0)$) is x_{n+m-1} . Similar to B_i suppose that the indices of x_i ($i =$

$0, 1, \dots, n+m-1$) are modulo $n+m$. This means that x_i and x_j be the same if i and j are equal mod $n+m$:

$$\dots, x_0, x_1, \dots, x_{n+m-1}, x_0, x_1, \dots, x_{n+m-1}, x_0, x_1, \dots, x_{n+m-1}, \dots$$

Let $U_i = \frac{q^i}{q^{n+m}-p^m}$, where $i = \overline{0, n+m}$. Suppose that oposite integers (a, b) , are such that $p^a U_0 \pm U_b \in \mathbb{Z}$ or, $p^a U_i \pm U_{i+b} \in \mathbb{Z}$ ($0 \leq i, 0 \leq b, i+b \leq n+m$). Analogously, suppose that for positive integers (a', b') , $p^{a'} U_{b'} \pm U_0 \in \mathbb{Z}$ or $p^{a'} U_{i+b'} \pm U_i \in \mathbb{Z}$ ($0 \leq i, 0 \leq b, i+b \leq n+m$). Suppose σ_s^r , where $s \leq r$, be the total number of S functions in the part $B_s \circ B_{s+1} \circ \dots \circ B_{r-1}$ of P . For example, $\sigma_s^s = 0$, since it is an empty part of P .

III. MAIN RESULTS

Theorem 1. $p^a x_i \pm p^{\sigma_i^{i+b}} x_{i+b} \in \mathbb{Z}$ ($0 \leq i, 0 \leq b, i+b \leq n+m$).

Theorem 2. $p^{a'+\sigma_i^{i+b'}} x_{i+b'} \pm x_i \in \mathbb{Z}$ ($0 \leq i, 0 \leq b', i+b' \leq n+m$).

Note: We will prove the last theorem. The first theorem is proved analogously.

Proof. It is first proved that the proposition holds true for one special P made of S_0 and T only. Then it is proved that if S_k in P is replaced by S_{k+1} , or S_{k-1} then the proposition remains true. The same for the replacement by S_{k+2} or S_{k-2} , etc. If all S_k in P are S_0 , then $x_i = 0$ for any i . Let x_i ($0 \leq i \leq n+m$) correspond to P_1 , whereas P_2 is the same with P_1 , except the first S_k in P_1 is now S_{k-1} . Let x'_i ($0 \leq i \leq n+m$) correspond to P_2 . Suppose $R_i = B_{n+m-i} \circ \dots \circ B_{n+m-2} \circ B_{n+m-1}$ is the fragment of i functions of P_1 from its right end. Designate by $R_i[b]$ the result R_i to b . For example, $R_0[b] = b$. Note that $P_1 = S_0 \circ R_{n+m-1}$ and $P_2 = S_{-1} \circ R_{n+m-1}$. Note that $P_1[x_0] = x_0$, $P_2[x'_0] = x'_0$, and

$$x_0 = \frac{pR_{n+m-1}[x_0] + k}{q}, x'_0 = \frac{pR_{n+m-1}[x'_0] + k - 1}{q}.$$

So, find the difference is

$$x'_0 - x_0 = \frac{pR_{n+m-1}[x'_0 - x_0] - 1}{q} = \frac{p^m(x'_0 - x_0)}{q^{n+m}} - \frac{1}{q}.$$

For $x'_0 - x_0$, we can write

$$x'_0 = x_0 - U_{n+m-1}.$$

Therefore

$$\begin{aligned} x'_{n+m-1} &= B_{n+m-1}(x'_{n+m}) = B_{n+m-1}(x'_0) = \\ &= B_{n+m-1}(x_0 - U_{n+m-1}) = \\ &= x_{n+m-1} - p^{\sigma_{n+m-1}^{n+m}} \cdot U_{n+m-2}. \end{aligned}$$

Analogously,

$$\begin{aligned} x'_{n+m-2} &= B_{n+m-2}(x'_{n+m-1}) = \\ &= B_{n+m-2}(x_{n+m-1} - p^{\sigma_{n+m-1}^{n+m}} \cdot U_{n+m-2}) = \\ &= x_{n+m-2} - p^{\sigma_{n+m-2}^{n+m}} \cdot U_{n+m-3}. \end{aligned}$$

In the same way,

$$x'_{n+m-k} = x_{n+m-k} - p^{\sigma_{n+m-k}^{n+m}} \cdot U_{n+m-k-1},$$

for $k = \overline{0, n+m-1}$. For example, if $k = n+m-1$ then $x'_1 = x_1 - p^{\sigma_1^{n+m}} \cdot U_0$. Using $\sigma_1^{n+m} = m-1$, we find $x'_1 = x_1 - p^{m-1} \cdot U_0$. More generally: $x'_i = x_i - p^{\sigma_i^{n+m}} \cdot U_{i-1}$, where $i = \overline{1, n+m}$. Since B_0 is not T ,

$$\begin{aligned} x'_0 &= B_0(x'_1) = B_0(x_1 - p^{m-1} \cdot U_0) = x_0 - \frac{1}{q}(p^m \cdot U_0 + 1) \\ &= x_0 - U_{n+m-1}, \end{aligned}$$

$$\text{or } x'_{n+m} = x_{n+m} - U_{n+m-1}.$$

Consider nonnegative integers (a', b') , such that $p^{a'} U_{b'} + U_0 \in \mathbb{Z}$. Let $p^{a'+\sigma_i^{i+b'}} x_{i+b'} + x_i \in \mathbb{Z}$ for x_i ($0 \leq i \leq n+m$). This true also for x'_i ($0 \leq i \leq n+m$). Indeed, if $i > 0$ then

$$\begin{aligned} &p^{a'+\sigma_i^{i+b'}} \cdot x'_{i+b'} + x'_i \\ &= p^{a'+\sigma_i^{i+b'}} \cdot (x_{i+b'} - p^{\sigma_{i+b'}^{n+m}} \cdot U_{i+b'-1}) \\ &\quad + (x_i - p^{\sigma_i^{n+m}} \cdot U_{i-1}) = \\ &= (p^{a'+\sigma_i^{i+b'}} \cdot x_{i+b'} + x_i) \\ &\quad - p^{\sigma_i^{n+m}} \cdot (p^{a'} \cdot U_{i+b'-1} + U_{i-1}) \in \mathbb{Z}. \end{aligned}$$

If $i = 0$ then

$$\begin{aligned} &p^{a'+\sigma_0^{b'}} x'_{b'} + x'_0 \\ &= p^{a'+\sigma_0^{b'}} \cdot (x_{b'} - p^{\sigma_{b'}^{n+m}} \cdot U_{b'-1}) \\ &\quad + (x_0 - U_{n+m-1}) = \end{aligned}$$

$$= (p^{a'+\sigma_0^{b'}} \cdot x_{b'} + x_0) - (p^{a'+\sigma_0^{n+m}} \cdot U_{b'-1} + U_{n+m-1}),$$

where $p^{a'+\sigma_0^{b'}} \cdot x_{b'} + x_0 \in \mathbb{Z}$. Also,

$$\begin{aligned} &p^{a'+\sigma_0^{n+m}} \cdot U_{b'-1} + U_{n+m-1} \\ &= p^{a'+m} \cdot U_{b'-1} + \frac{1}{q}(p^m \cdot U_0 + 1) \\ &= \frac{p^m(p^{a'} \cdot U_{b'} + U_0) + 1}{q} \\ &= \frac{p^{m+a'} \cdot q^{b'} + q^{n+m}}{q \cdot (q^{n+m} - p^m)} \in \mathbb{Z}, \end{aligned}$$

because $p^{a'} \cdot U_{b'} + U_0 \in \mathbb{Z}$ and $q|p^{m+a'} \cdot q^{b'} + q^{n+m}$.

If S_k is changed to S_{k+1} the proof is similaris considered in a similar way. By induction principle the proof is complete.

IV. EXAMPLES

Let us take $p = 5$. Consider the composition $P = B_0 \circ B_1 \circ B_2 \circ B_3 \circ B_4 \circ B_5 \circ B_6 = S_2 \circ T \circ T \circ S_3 \circ T \circ T \circ S_{-1}$. Here $m = 3$, $n = 4$ and $2^{m+n} - 5^m = 3$. From $S_2 T T S_3 T T S_{-1}(x) = x$ we find $x_0 = \frac{223}{3} = x_7$. The remaining terms $x_1 = x_8 = \frac{88}{3}$, $x_2 = \frac{176}{3}$, $x_3 = \frac{352}{3}$, $x_4 = \frac{139}{3}$, $x_5 = \frac{278}{3}$, $x_6 = \frac{556}{3}$, are determined through

$$T \circ T \circ S_3 \circ T \circ T \circ S_{-1} \circ S_2(x) = x,$$

$$T \circ S_3 \circ T \circ T \circ S_{-1} \circ S_2 \circ T(x) = x,$$

$$S_3 \circ T \circ T \circ S_{-1} \circ S_2 \circ T \circ T(x) = x,$$

$$T \circ T \circ S_{-1} \circ S_2 \circ T \circ T \circ S_3(x) = x,$$

$$T \circ S_{-1} \circ S_2 \circ T \circ T \circ S_3 \circ T(x) = x,$$

$$S_{-1} \circ S_2 \circ T \circ T \circ S_3 \circ T \circ T(x) = x,$$

respectively. We also find the numbers $U_i = \frac{2^i}{3}$ ($i = 0, 1, \dots, 7$). Note that $5^1 \cdot U_0 + U_2 = 3$ is an integer. So, we can take $a = 1$, $b = 2$ in the plus case of Theorem 1. We observe that $5^1 \cdot x_i + 5^{\sigma_i^{i+2}} \cdot x_{i+2} \in \mathbb{Z}$ for each $i = 0, 1, \dots, 4$. Indeed,

$$5x_0 + 5^{\sigma_0^{2}} x_2 = 5 \left(\frac{223}{3} \right) + 5 \left(\frac{176}{3} \right) = 665,$$

$$5x_1 + 5^{\sigma_1^3} x_3 = 5 \left(\frac{88}{3} \right) + \left(\frac{352}{3} \right) = 264,$$

$$5x_2 + 5^{\sigma_2^4} x_4 = 5 \left(\frac{176}{3} \right) + 5 \left(\frac{139}{3} \right) = 525,$$

$$5x_3 + 5^{\sigma_3^5} x_5 = 5 \left(\frac{352}{3} \right) + 5 \left(\frac{278}{3} \right) = 1050,$$

$$5x_4 + 5^{\sigma_4^6}x_6 = 5\left(\frac{139}{3}\right) + 5\left(\frac{556}{3}\right) = 417,$$

$$5x_5 + 5^{\sigma_5^7}x_7 = 5\left(\frac{278}{3}\right) + 5\left(\frac{223}{3}\right) = 835,$$

$$5x_6 + 5^{\sigma_6^8}x_8 = 5\left(\frac{556}{3}\right) + 5\left(\frac{88}{3}\right) = 1660,$$

are all integers.

Let us take $p = 11, q = 5$. Let us take composition $P = B_0 \circ B_1 \circ B_2$, where $B_0(x) = T(x) = \frac{x}{5}$, $B_1(x) = S_4(x) = \frac{11x+4}{5}$, $B_2(x) = S_{-1}(x) = \frac{11x-1}{5}$. Here $m = 2$, $n = 1$ and $5^{m+n} - 11^m = 4$. The solution of the linear equation $B_0 \circ B_1 \circ B_2 = T \circ S_4 \circ S_{-1}(x) = x$ is $x_0 = \frac{9}{4} = x_3$. The remaining terms $x_1 = \frac{45}{4}$, $x_2 = \frac{19}{4}$, can be determined by solving the equations $S_4 \circ S_{-1} \circ T(x) = x$, $S_{-1} \circ T \circ S_4(x) = x$, respectively. We also find the numbers $U_i = \frac{2^i}{3}$ ($i = 0, 1, 2, 3$). Note that $11^1 \cdot U_0 + U_1 = 3$ is an integer. So, we can take $a = 1, b = 1$ in the plus case of Theorem 1. Again, we observe that $11^1 \cdot x_i + 11^{\sigma_i^{i+1}} \cdot x_{i+1} \in \mathbb{Z}$, for each $i = 0, 1, 2, 3$. Indeed,

$$11^1 \cdot x_0 + 11^{\sigma_0^1} \cdot x_1 = 11^1 \cdot \left(\frac{9}{4}\right) + 11^0 \cdot \left(\frac{45}{4}\right) = 36,$$

$$11^1 \cdot x_1 + 11^{\sigma_1^2} \cdot x_2 = 11^1 \cdot \left(\frac{45}{4}\right) + 11^1 \cdot \left(\frac{19}{4}\right) = 176,$$

$$11^1 \cdot x_2 + 11^{\sigma_2^3} \cdot x_3 = 11^1 \cdot \left(\frac{19}{4}\right) + 11^1 \cdot \left(\frac{9}{4}\right) = 77,$$

are all integers.

Similar examples can be given for minus case of Theorem 1, and both plus and minus cases of Theorem 2. The given examples agree perfectly the mentioned findings of this paper.

V. ALGORITHM IN PASCAL

The Pascal language program [2] (<https://rextester.com/TDV58751>) written by the author describes the algorithm of division by two in quinary numeral

system. Before running the program one can change $P := '412422033201303'$ two another number P written in quinary numeral system and calculate its half. Using this program it is calculated that half of the number P is 203433241323124.

It is possible to write similar programs for division by any number q of any p -adic number.

VI. CONCLUSION

The results of the current paper generalize the results given in [1,3-6]. The main novelty is that the denominators in S_k and T functions are not just 2 but any number q relatively prime to p . The worked out examples and the algorithm as a Pascal program demonstrated some applications of these theoretical results.

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